

Domination Parameters and Added Matchings

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Abstract

We consider the augmentation problem of how domination parameters behave when a perfect matching P of the complement is added to the graph. We focus on the case that the graph is a tree, and inter alia show that if T is a tree of even order n that is not a star, then $T + P$ has domination number at most $2n/5$, independent domination number at most $n/2 - 1$, and total domination and upper domination number at most $n/2$. Further, there exists a choice of P such that $T + P$ has total domination number at most $n/3$. All these bounds are sharp.

1 Introduction

We consider the problem of how domination parameters behave when a matching is added to the graph. This can be thought of as an augmentation type problem. For example, Cropper et al. [4] considered the domination number of hamiltonian 3-regular graphs, or equivalently, adding a perfect matching to a cycle. But there are other questions about augmentation. For example, several authors considered domination in “generalized prisms”: these are graphs obtained from two copies of a base graph by adding a perfect matching between the copies. See for example [12, 15].

We focus here on the question: Given a family $\mathcal{F}$ of even-order graphs and a parameter ψ ,

What is the extremum value of $\psi(G + P)$ over all $G \in \mathcal{F}$ of given order and perfect matchings P in the complement of G ?

This question comes in four variations: $\max_G \max_P$, $\max_G \min_P$, $\min_G \max_P$, and $\min_G \min_P$. We call these problems MM, MM, MM, and MM respectively (though in some cases the problem is trivial).

Our focus is on the case when $\mathcal{F}$ is some collection of sparse graphs, since then the added perfect matching has the most impact. In particular, the problem in the set of trees is interesting and that is what we spend the most time on. But for some of the problems, we also consider (regular) graphs of small minimum degree. Similar ideas but for removing a matching were studied in [2, 3]. The topic could also be studied for more general matchings—though in some cases some maximality condition is needed to avoid the empty set being the solution—but that is not explored here.

In Section 2 we consider ordinary domination, and show that if T is a tree of even order n that is not a star, then for any perfect matching P in the complement, $T + P$ has domination number at most $2n/5$, and that the question for the best P is related to the notion of partial domination. We also observe some bounds for $G + P$ for graphs G with small minimum degree. In Section 3 we show that $T + P$ has total domination number at most $n/2$, but for every tree there is a choice of P such that the total domination number of $T + P$ is at most $n/3$. In Section 4 we show that $T + P$ has independent domination number at most $n/2 - 1$. Similar questions are considered for independence number in Section 5 and for upper domination and upper total domination number in Section 6. For more information on the domination parameters involved, see for example [7].

2 The Domination Number

Most of our results are for the case that $\mathcal{F}$ is the set of trees. We will frequently make use of the following (known) fact. (One can readily prove this by induction using the observation that any tree of order at least 6 has a pair of end-vertices such that removing them leaves a tree that is not a star.)

Lemma 1 *If T is a tree/forest of even order that is not a star, then its complement $\overline{T}$ contains a perfect matching.*

The first question we consider is Problem MM for trees. That is, we ask what is the maximum domination number of a tree plus a perfect matching? We observe that the answer is $2n/5$. We denote by $\gamma(G)$ the domination number of a graph G .

Theorem 2 *If T is a tree of even order n that is not a star and P a perfect matching in $\overline{T}$, then*

$$\gamma(T + P) \leq \frac{2}{5}n,$$

and this bound is sharp.

PROOF. The graph $H = T + P$ is connected and has minimum degree two. Blank [1] and McCuaig and Shepherd [11] showed that such a graph with $n \geq 8$ has domination number at most $2n/5$. In the latter paper it is shown that the only even-order graph with domination number more than $2n/5$ is C_4 , but one cannot build C_4 by adding a perfect matching to a tree. $\square$

To show that the bound of Theorem 2 is sharp, we use the extremal family determined by McCuaig and Shepherd [11]. Specifically let k be even. Then let $F_1, \dots, F_k$ be disjoint copies of the tree of order 5 and diameter 3, and let T_k be the tree obtained by joining, for $1 \leq i < k$, the vertex a_i of degree 2 in F_i to the vertex b_{i+1} of degree 3 in F_{i+1} . Then a suitable matching P of T_k is obtained by joining each a_i with a neighbor of b_i in F_i , joining the other two end-vertices in F_i , and joining b_{2i-1} with b_{2i} for $1 \leq i \leq k/2$. Figure 1 provides a picture of T_4 and the perfect matching. It is easily checked that one needs two vertices from each F_i in the dominating set, and indeed, that the resultant graph is in the extremal family provided in [11].

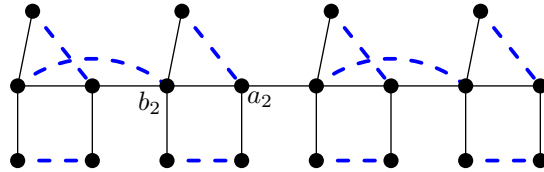


Figure 1: The tree T_4 and an augmenting perfect matching

We note though, that the trees T_k have vertices of degree 4. This suggests the question of what happens if one starts with a tree with maximum degree at most 3. There are several trees of order 10 where the answer is 4. But there is none of order $n \geq 20$: it can be checked that, for the extremal graphs of these orders from [11], removing any perfect matching disconnects the graph.

Perhaps the asymptotics is $\frac{3}{8}n$? We have examples of orders 16 and 24 where $\gamma(T + P) = \frac{3}{8}n$, but unclear thereafter.

If one increases the minimum degree of a graph in $\mathcal{F}$ to two, we know the answer to Problem MM, namely $3n/8$. Here the resultant graph $H = G + P$ has minimum degree three, and Reed [13] showed that such graphs have domination number at most $3n/8$. Further, we observe that there are extremal graphs with the desired property. Let J be the cubic graph on 8 vertices formed by one Y - Δ operation applied to $K_{3,3}$. Then a graph H achieving domination number $3n/8$ can be constructed by taking multiple disjoint copies of J and choosing one vertex from the triangle in each copy and joining up the chosen vertices arbitrarily. (See Figure 6.10 in [7].) We note here that J has a hamilton cycle C , and thus the removal from H of the matching P consisting of the edges not in any C leaves a connected graph.

We consider next when the set $\mathcal{F}$ is the set of 2-regular graphs. The best we can do here is to again use the known results for the resultant graphs. Kostochka and Stodolsky [10] provided an upper bound for the domination number of cubic graphs of $\frac{4}{11}n$ for $n > 8$. And in [9] they provided some constructions of cubic graph with large domination number. (We note that the main construction of that paper does not yield a graph with a perfect matching, but at the end they provide a construction for cubic graphs that do have a perfect matching and whose domination number is asymptotically bigger than $n/3$.)

And finally, if we consider $\mathcal{F}$ to be the set of all connected 2-regular graphs, in other words only $\{C_n\}$, then the question was resolved by Cropper et al. [4]. They showed that the maximum value of $\gamma(C_n + P)$ over all perfect matchings P in the complement is $\lfloor (n+1)/3 \rfloor$ for even n . Note that in particular this means that for even-length cycles with order congruent to 0 or 2 mod 3, it is possible to add a perfect matching and not decrease the domination number from its value of $\lceil n/3 \rceil$. This suggests the side question of: what is the maximum domination number of a graph such that one can add a perfect matching to it and the domination number is unchanged?

We consider Problem MM next. This problem is related to *partial domination* (see for example [5]). Recall that $N_G[S]$ denotes the closed neighborhood of S : the set consisting of S and its neighbors.

Theorem 3 *For a graph G and integer s , let $\pi_G(s)$ be the maximum of $|N_G[S]|$ over all sets S of size s . Then, the minimum value of $\gamma(G + P)$ over all perfect matchings P in $\overline{G}$ is at least the minimum s such that $\pi_G(s) \geq n - s$. Furthermore, the minimum is attained whenever G is a tree.*

PROOF. It is immediate that for any s , if S is any dominating set of $G + P$ of size s , then $|N_G[S]| \geq n - s$, since removing the perfect matching P from $G + P$ can reduce the number of vertices dominated by S by at most $|S|$. Thus $\pi_G(s) + s \geq n$ for all s for which there exists a dominating set.

It remains to show attainment for trees. That is, given that there exists a set S of vertices of size s in tree T such that $|N_T[S]| + s \geq n$, then $\gamma(T + P) \leq s$ for some choice of perfect matching P in the complement $\overline{T}$.

Out of all sets S of size s , let S have the maximum $|N_T[S]|$, and subject to this, let S have the maximum number of edges leaving S , that is, maximum $|[S, V - S]|$. Let J be the set of undominated vertices $V(T) - N_T[S]$. Consider a subset S_0 of S with $|S_0| = |J|$. If the complement of $T - (S_0 \cup J)$ has a perfect matching, then one can extend it by pairing up S_0 and J arbitrarily to form a perfect matching P of $\overline{T}$ where S dominates $T + P$. So (by Lemma 1) assume $T - (S_0 \cup J)$ is a star, say with center v .

Suppose $v \notin S$. Then v is dominated by some vertex $w \in S$. By the lack of cycles, the vertex w has no common neighbor with v and thus has no neighbor outside S except v . It follows that $(S - \{w\}) \cup \{v\}$ dominates all of $V(T) - J$ and has more edges leaving it, which contradicts the maximality of $|[S, V - S]|$. Hence $v \in S$.

It follows that $|S| = |J| + 1$, since any other vertex of $S - S_0$ is dominated by v , and so it can be replaced in S by a vertex of J , contradicting the maximality of $|N_T[S]|$. Now consider a subset S_1 of S of size $|J|$ that contains v . Again we are done, unless the subgraph $T - (S_1 \cup J)$ is a star, say with center v' . But then vertices v and v' have common neighbors creating a cycle, unless $T - (S_1 \cup J)$ has only two vertices. But in that case, the subgraph $T - (S_1 \cup J)$ is a K_2 , and both vertices can be viewed as the center of the star, and hence by the above reasoning are in S . This is a contradiction. Hence we have shown that for some choice of S of size s there exists a perfect matching P in $\overline{T}$ such that S dominates

$T + P$. $\square$

We do not know the answer to Problem MM for trees. There are many trees T where the minimum value of $\gamma(T + P)$ over all perfect matchings P in $\bar{T}$ is $\lceil n/4 \rceil$. For example, a *corona* $H \circ K_1$ of a graph H is defined as the graph that results by adding an end-vertex adjacent to each vertex of H . Such graphs have maximum domination number. But by taking a minimum dominating set of H and if necessary adding vertices of H to form a set S of size $n/4$ (where n is the order of $H \circ K_1$), it follows by Theorem 2 that $\pi_{H \circ K_1}(n/4) \geq 3n/4$. Indeed the corona has a perfect matching such that the result has domination number at most $n/4$. It is the case that, for small n , the value $\lceil n/4 \rceil$ is best possible. However, there are trees where the value is larger.

The largest value required that we know of asymptotically is the following construction. Let B_m be a graph of order $n = 5m$ obtained from m disjoint copies of the path P_5 on five vertices by adding edges between the central vertices arbitrarily to form a tree. It can be shown that the augmentation problem here solves to $4n/15$ when m is a multiple of 3. See Figure 2 for an example, where the 8 ringed vertices dominate all but 8 vertices; adding a perfect matching P in the complement that matches the ringed vertices to the undominated vertices shows that $\gamma(B_m + P) \leq 4n/15$. At the same time, an upper bound of $\lceil n/3 \rceil$ for the best perfect matching follows from the result on total domination (Theorem 5). We can improve that bound slightly, but cannot match the $4n/15$ and thus omit the details.

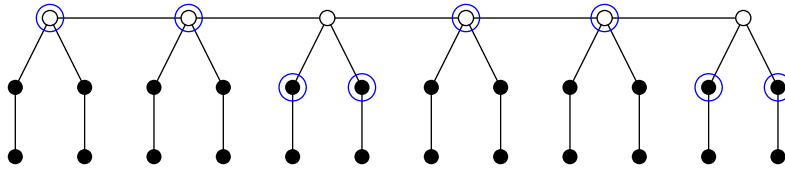


Figure 2: The tree B_4 and a partial dominating set

Problem MM for the even cycle was answered by Cropper et al. [4] to be $\lceil n/4 \rceil$. It is not hard to extend this result to 2-regular graphs in general: that is, given any collection of disjoint cycles of even total order, it is possible to add a perfect matching such that the result has domination number $\lceil n/4 \rceil$ (the smallest

possible for a 3-regular graph).

3 The Total Domination Number

We turn our attention to total domination. We denote by $\gamma_t(G)$ the total domination number of graph G ; that is, the minimum size of a dominating set that induces a subgraph with no isolates. We start with Problem MM for trees. That is, what is the maximum total domination number of a tree T plus a perfect matching P in $\bar{T}$?

Theorem 4 *If T is a tree of even order n that is not a star and P a perfect matching in $\bar{T}$, then*

$$\gamma_t(T + P) \leq \frac{1}{2}n,$$

and this bound is sharp.

PROOF. Let $H = T + P$. We claim H cannot contain a path on three vertices all of whose vertices have degree 2 in H . For suppose there is such a path $v_1v_2v_3$. Without loss of generality, the edge incident with v_2 that is in P is v_2v_1 . Then the other edge incident with v_3 is also in P . But that means T contains a K_2 -component consisting of the edge v_2v_3 , a contradiction.

It is known (see [8]) that if H is a graph of order n with $\delta(H) = 2$ such that the set of vertices of degree 2 in H induce a subgraph with maximum degree at most 1, then $\gamma_t(H) \leq \frac{1}{2}n$. Hence in our case, we infer that $\gamma_t(T + P) = \gamma_t(H) \leq \frac{1}{2}n$. $\square$

The bound in Theorem 4 is achieved, for example, as follows. Let T' be an arbitrary tree and let $T = T' \circ P_3$; that is, T is the graph of order $4|V(T')|$ obtained from T' by attaching a path of order 3 to each vertex of T' so that the resulting paths are vertex-disjoint. Let P be the perfect matching in $\bar{T}$ that joins every end-vertex in T to the vertex at distance 2 from it and joins every neighbor of an end-vertex to the vertex at distance 2 from it in T . For example, if $T' = P_4$, then $T = T' \circ P_3$ is illustrated in Figure 3(a) and the associated perfect matching P in $\bar{T}$ is indicated by the blue edges in Figure 3(b). We claim that the resultant

graph $H = T + P$ has $\gamma_t(H) = n/2$ where n is its order. The upper bound is from the Theorem 4. For the lower bound, we note that in order to totally dominate the vertices of degree 2 in H , every total dominating set of H contains at least one neighbor of each such vertex. Moreover, in order to totally dominate the neighbors of the vertices of degree 2 in H , at least one additional vertex within distance 2 from each vertex of degree 2 is needed.

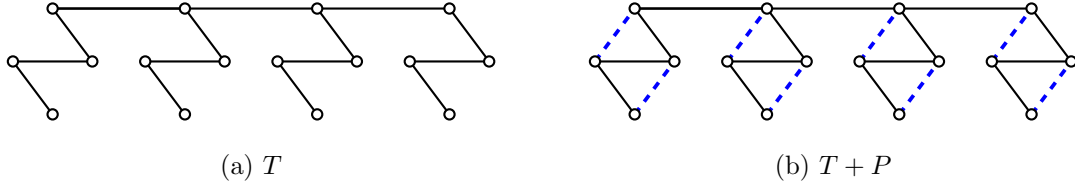


Figure 3: A tree T with associated perfect matching P in $\bar{T}$

If in Theorem 4 one replaces “tree” by “connected graph with $\delta(G) \geq 2$ ”, the bound does not change. The upper bound follows from taking a spanning tree and applying the above theorem. Further, there are two infinite families of cubic graphs with total domination number half their order, introduced in [6]. It is easily checked that these graphs have hamiltonian cycles, and thus can be produced by adding a perfect matching to C_n .

We next consider Problem MM for trees.

Theorem 5 *Let T be a tree of even order $n \geq 4$ that is not a star. Then there exists a set $D \subseteq V(T)$ of size at most $\lceil n/3 \rceil$, and a perfect matching P in $\bar{T}$, such that D is a total dominating set of $T + P$.*

The theorem is proved below. The sharpness is immediate from T being the path. In that case, for any matching P the graph $H = T + P$ has maximum degree 3, and so $\gamma_t(H) \geq n/3$.

3.1 Proof of Theorem 5

For the proof of Theorem 5 it is helpful to allow graphs of odd order. So we prove the following generalization. Recall that a *near-perfect matching* is one that

leaves exactly one vertex unmatched.

Theorem 6 *Let T be a tree of order $n \geq 4$ that is neither a path where n is an odd multiple of 3 nor a star where n is even. Then there exists a set $D \subseteq V(T)$ of size at most $\lceil n/3 \rceil$, and a perfect/near-perfect matching P in $\overline{T}$, such that D is a total dominating set of $T + P$.*

We start by considering trees of radius 2. Note that the bound in the following lemma is **round down** of $n/3$ rather than round up.

Lemma 7 *Let T be a tree of radius 2 with order $n \geq 6$ and center C . Then there exists a set $D \subseteq V(T)$ of size at most $\lfloor n/3 \rfloor$ containing C , and a perfect/near-perfect matching P in $\overline{T}$ where all of C is matched, such that D is a total dominating set of $T + P$.*

PROOF. If T has diameter 3, then C provides the desired D , and any perfect / near-perfect matching P that matches both vertices of C suffices.

So assume T has diameter 4. Let v be the unique central vertex and root the tree at v . Order the neighbors of v as $w_1, \dots, w_k$ such that the count $c_1, \dots, c_k$ of their children is non-increasing. By the diameter condition it holds that $k \geq 2$. Build the set D by taking v and $w_1, \dots, w_{d-1}$ stopping as soon as the number of vertices not dominated by D is at most the number of vertices in D . (That is, $d \geq \sum_{i=d}^k c_i$.) Since the diameter is four, there are at least two vertices nonadjacent to v , and thus $d > 1$.

Claim. *It holds that $n \geq 3d$.*

Proof of claim. First assume $w_k \notin D$. If $c_d \geq 2$, then we associate each $w_i \in D$ with its children (recall that $c_i \geq c_d$ for such vertices), and associate v with the children of w_d . Since all these vertices are distinct, it follows that $n \geq (c_d + 1)d \geq 3d$.

If $c_d = 1$, then we associate each $w_i \in D$ where $c_i \geq 2$ with its children. Further, we associate each $w_i \in D$ where $c_i = 1$ with its child and one undominated vertex. This is possible because if $c_{d-1} = 1$ then the fact that w_{d-1}

was needed for D implies that number of un-dominated vertices is at least $d - 1$. Finally we associate v with w_k and one other vertex not already associated. If this is not possible, then $w_{k-1} \in D$ and the child of w_k is already associated. But that means that $c_{k-1} = 1$, which means that $k \geq 3$ by the order constraint, and so we should have stopped at $d - 1 = k - 2$, a contradiction. It follows that $n \geq 3d$.

Second assume $w_k \in D$. The fact that one needs w_k implies that $c_k > k$. Hence $n \geq 1 + k(c_k + 1) \geq 1 + k(k + 2)$. Since $k \geq 2$, the latter quantity is at least $3(k + 1)$. Thus $n \geq 3d$. $\triangleleft$

By the claim, it follows that $|D| \leq \lfloor n/3 \rfloor$. One can pair off the vertices not dominated by D with a subset of D that includes v . The vertices not already paired off induce a disconnected forest. Hence one can extend the pairing off to a perfect/near-perfect matching P in $\bar{T}$ such that D is a total dominating set in $T + P$. $\square$

We will also need the following observation about exceptional paths:

Lemma 8 *If U is a path of order n an odd multiple of 3, then for any designated vertex z , there exists a set $D \subseteq V(U)$ of size $n/3$ and a near-perfect matching P in $\bar{U}$, such that the open neighborhood of D in $U + P$ is $V(U) - \{z\}$.*

PROOF. Break U into $n/3$ subpaths of order 3. Take for D the vertex z and the center of each subpath not containing z . Then for P , pair off the vertices of $D - \{z\}$ and add edges making each subpath a triangle. $\square$

We now prove Theorem 6. The proof is by induction on n . The result is immediate for odd-order stars (since $n \geq 4$). The trees of radius 2 are covered by Lemma 7. So assume tree T has radius more than 2. We define a **subtree rooted at vertex** v as any component containing v obtained from T by deleting one edge incident with v . We say that the subtree has **height** ℓ if the maximum distance from v to any vertex of the subtree is ℓ .

Case A: *There is a subtree H of height 2 with order at least 6 such that the tree $T' = T - H$ satisfies the inductive hypothesis.* Assume H is not a star. Let set D_H and matching P_H result from the above lemma applied to H , and set D'

and matching P' result from the inductive hypothesis applied to T' . If at least one of H and T' has even order, then let the perfect / near-perfect matching P of $\bar{T}$ be $P_H \cup P'$. If both H and T' have odd order, then the unmatched vertex in H is not the root v by Lemma 7. Thus one can form a perfect matching P of $\bar{T}$ from $P_H \cup P'$ by pairing off the two unmatched vertices in that case. Further, let $D = D_H \cup D'$. It follows that D is a total dominating set of $T + P$. And, since for any integers $0 < a < n$ it holds that $\lfloor a/3 \rfloor + \lceil (n-a)/3 \rceil \leq \lceil n/3 \rceil$, the set D has the desired size.

If H is a star, then it is not hard to show that one can take D_H to consist of the root of H and its neighbor, and amend and extend P' to a perfect / near-perfect matching P in $\bar{T}$ such that $D_H \cup D'$ is a total dominating set of $T + P$ of the desired size.

Case B: *There is a subtree H of height 2 with order at least 6 but $T' = T - H$ does not satisfy the hypothesis of the theorem.* Assume H is not a star. (If H is a star one can proceed similarly.) Apply Lemma 7 to H to yield set D_H and matching P_H . Since T has radius more than 2, the tree T' has at least three vertices. There are two possibilities.

(i) Assume T' is an even-order star, necessarily of order at least 4. The tree T' has a total dominating set D' of order 2. Then form $D = D_H \cup D'$. Further, note that P_H includes an edge e joining two vertices of H neither of which is in D_H . It follows that D is a total dominating set of $T + (P_H - e)$, while $P_H - e$ can be extended to a perfect/near-perfect matching P in $\bar{T}$ by starting by pairing the ends of e with D' . Furthermore, $|D| \leq \lceil n/3 \rceil$, as required.

(ii) Assume T' is a path with order m an odd multiple of 3. Note that the root v of H is in D_v and has a neighbor in T' , say z . By Lemma 8 there exists a set D' of T' of size $m/3$ and near-perfect matching P' in $\bar{T}'$ such that the open neighborhood of D' in $T' + P'$ is $V(T') - \{z\}$. Then form matching P from $P_H \cup P'$ by joining the unmatched vertex of H if there is one (which by Lemma 7 is not v) to the unmatched vertex of T' . It follows that $D = D_H \cup D'$ is a total dominating set of $T + P$ of the desired size.

Case C: *All subtrees of height 2 have order at most 5.* Note that since T has radius more than two, there are at least two subtrees of height 2. Consider two such subtrees, say H_1 and H_2 with orders n_1 and n_2 where $n_1 \geq n_2$.

Case C1: $n_2 \geq 4$. Note that $n_1 + n_2 \leq 10$. If $n \leq 11$ the result can easily be checked. So assume $n \geq 12$. Let Q be any set of $12 - (n_1 + n_2)$ vertices of $T - (H_1 \cup H_2)$ such that removing Q from it leaves a connected graph (or one with no vertices at all), and let $R = V(H_1) \cup V(H_2) \cup Q$. Note that $|R| = 12$. Assume one can apply the induction hypothesis to tree $T' = T - R$ (or it has no vertex). Then one obtains a set D' and matching P' . Let D_Q contain the root vertex and one largest-degree neighbor from each H_i . Note that $|D_Q| = 4$. Furthermore, the set D_Q dominates at least eight vertices of R (including itself). Thus one can pair off (a subset of) D_Q with the undominated vertices in R , and extend to a perfect matching P_R in the complement of $T[R]$, so that D_Q is a total dominating set of $T[R] + P_R$. Finally, let $D = D' \cup D_Q$ and $P = P' \cup P_R$, and the statement of the theorem follows.

If the tree T' does not satisfy the induction hypothesis, then there are three cases. The case that T' has just one vertex is easily handled. Consider the case that T' is an even-order star. If T' has order at least 4, then it has a suitable set D' but only a matching P' that leaves two vertices unmatched. But P_R has an edge joining two vertices not in D_Q , and one can replace that edge by two edges joining the ends to the unmatched vertices in T' . If T' has order 2, then one of its vertices, say z , is adjacent to D_Q and adding z to D_Q provides a suitable total dominating set, and the perfect matching P can be resolved as before. Consider the case that T' is a path with order an odd multiple of 3. Then let z be the neighbor of the root vertex of H_1 in T' . Apply Lemma 8 to T' to produce set D' and matching P' . If we let $D = D_Q \cup D'$ and $P = P_R \cup P'$, it follows that P is a near-perfect matching in $\bar{T}$ and D is a total dominating set of $T + P$ of the desired size.

Case C2: $n_2 = 3$. If $n_1 = n_2 = 3$, then let $Q = V(H_1) \cup V(H_2)$. Assume one can apply the inductive hypothesis to tree $T' = T - Q$. Then extend the set D' by adding the set D_Q being the central vertex from H_1 and H_2 , and extend the matching P' starting with the edge joining the two vertices of D_Q . The result satisfies the statement of the theorem.

If $n_1 > 3$ then let Q be $V(H_1)$ together with the $6 - n_1$ vertices of H_2 farthest from its root. Again assume one can apply the inductive hypothesis to tree $T' = T - Q$. Let D_Q be the set of the root of H_1 and one largest-degree

child from H_1 . Extend P' to a perfect / almost perfect matching of $\bar{T}$ starting with edges joining the two vertices of D_Q to the two vertices of Q not already dominated. The set $D' \cup D_Q$ is a total dominating set of $T + P$ of the desired size.

The case where T' is an even-order star is handled as before. Likewise, the case where T' is a path of order an odd multiple of 3 and D_Q has a neighbor in T' is handled as before. So it remains to consider the case that $n_1 = n_2 = 3$ and T' is a path of order m an odd multiple of 3.

Note that in this case n is itself an odd multiple of 3 and so T is not a path. Assume at least one of H_1 or H_2 is joined to an interior vertex z of T' . Then it can readily be argued that there is a subset $D' \subseteq V(T')$ of size $m/3$ containing z whose open neighborhood in T' has size $2m/3$ and hence whose open neighborhood in T has size more than $2m/3$. Form D from D' by adding the centers of both H_1 and H_2 . It follows that one can add a partial matching inside $T[D]$ that covers all the isolated vertices of D (and possibly one non-isolate), and pair off (a subset of) what remains of D with the undominated vertices, and extend to a near-perfect matching P of T , so that D is a total dominating set of $T + P$. Finally assume both H_1 and H_2 are joined to an end vertex z of T' (necessarily the same since T is not a path). Then one can build a suitable D starting with z and its two neighbors in H_1 and H_2 , and proceed as before.

This completes the induction and proof of Theorem 6. Theorem 5 follows.

4 Independent Domination

We consider independent domination next. Let $i(G)$ denote the minimum size of an independent dominating set. It is immediate that one can take at most one vertex from each edge of a perfect matching P , and so $i(G + P) \leq n/2$. But this value is not quite attainable in trees.

Theorem 9 *If T is a tree of even order n that is not a star, then for every perfect matching P in $\bar{T}$ it holds that*

$$i(T + P) \leq n/2 - 1,$$

and this is best possible.

PROOF. Let P be a perfect matching in $\overline{T}$. Assume first that there exists an edge xy of P joining two vertices x and y at distance two in T . Then build a maximal independent set S of $H = T + P$ starting with the common neighbor of x and y . The set S contains neither x nor y , and at most one vertex from every other edge of P , and thus has size at most $n/2 - 1$. Hence $i(H) \leq n/2 - 1$.

Assume second that there is no edge of P joining vertices at distance two in T . For each edge $e \in P$, let $\sigma(e)$ be the sum of the degrees in T of its ends. Note that the sum $\sum_e \sigma(e)$ equals the degree sum of T . Thus the average $\sigma(e)$ is $2(n-1)/(n/2) = 4 - 4/n$. Let $e = xy$ be an edge of P with maximum $\sigma(e)$. If $n = 4$, then the theorem is immediate; so assume otherwise, whence $\sigma(e) > 3$. Let $X = N_T(x) - \{y\}$ and $Y = N_T(y) - \{x\}$. Since T is acyclic, there is at most one edge of T joining X to Y . Since $|X| + |Y| > 3$, it follows that either one set has size at least 3, or both sets have size 2, and in either case P cannot account for all the remaining possible edges joining X to Y . That is, there exist vertex $x' \in X$ and vertex $y' \in Y$ such that they are not adjacent in H . Now, build a maximal independent set S of H starting with x' and y' . Again this set contains neither x nor y , and at most one vertex from every other edge of P , and thus has size at most $n/2 - 1$. $\square$

There are many trees T where there exists a perfect matching P in the complement such that $i(T + P) = n/2 - 1$. For example, one such tree is the double-star with central vertices $\{u, v\}$ having equal degree: choose P to contain $n/2 - 2$ edges joining $N(u) - \{v\}$ to $N(v) - \{u\}$. On the other hand, the theorem does not generalize to all graphs. If one starts with the graph G that is $K_{m,m}$ minus a perfect matching P , then adding back the perfect matching yields a graph $G + P$ with independent domination number half its order.

We turn to Problem MM. We do not know the answer. The largest value we know of asymptotically is $\frac{2}{7}n$.

Theorem 10 *There exists infinitely many trees T of even order n such that for all perfect matchings P of $\overline{T}$ it holds that $i(T + P) > \frac{2}{7}n$.*

PROOF. For even b define tree J_b as follows. Start with a star with center v on $3b$ edges and subdivide each edge. Then take a star with center w on b edges. Finally, add an edge joining v and w . The resultant graph has $7b + 2$ vertices. Figure 4 shows J_2 .

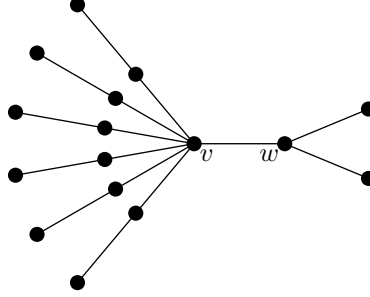


Figure 4: The tree J_2

We claim that for any perfect matching P in the complement of J_b it holds that $i(J_b + P) \geq 2b + 1$. To see this, first consider building an independent dominating set of $H = J_b + P$ starting with v . Then $J_b - N_{J_b}[v]$ is a graph of $4b$ isolates. Thus $H - N_H[v]$ has maximum degree at most one and at least $4b - 1$ vertices, and so its domination number is at least $\lceil (4b - 1)/2 \rceil = 2b$. Similarly, consider building an independent dominating set of H starting with w . Then $J_b - N_{J_b}[w]$ is a 1-regular graph of order $6b$. Thus $H - N_H[w]$ has maximum degree at most two and at least $6b - 1$ vertices, and so its domination number is at least $\lceil (6b - 1)/3 \rceil = 2b$. Finally, if neither v nor w is used for the independent dominating set of H , then one needs at least $n/3$ vertices. $\square$

5 Independence Number

In this section we consider the independence number, which we denote by $\alpha(G)$. Any graph with a perfect matching has independence number at most half its order. Perhaps surprisingly, this value is the answer for trees for both Problems MM and mM:

Theorem 11 *If T is a tree of even order n that is not a star, then there exists*

a perfect matching P in $\bar{T}$ such that

$$\alpha(T + P) = n/2,$$

and this is best possible.

PROOF. It is immediate that $\alpha(T + P) \leq n/2$ for any such P . It suffices to show that there exists an independent set A of T of size $n/2$ and a perfect matching P in $\bar{T}$ that matches A with $V(T) - A$. For, from this it follows that the set A is independent in $T + P$ and so $\alpha(T + P) = n/2$.

Assume first that the maximum degree of T is less than $n/2$. Consider any independent set A in T of size $n/2$ and let $B = V(T) - A$. We will apply Hall's Theorem to show the existence of the perfect matching P in $\bar{T}$ that matches A with B .

Consider a nonempty subset A' of A , and let $c(A')$ be the number of vertices of B in the neighborhood of A' in $\bar{T}$. Assume $|A'| = 1$, say $A' = \{a\}$. Then there must be at least one vertex in B that a is not adjacent to in T , since a has degree less than $|B|$, and so $c(A') \geq 1$. Assume $2 \leq |A'| \leq n/2 - 1$. Say A' contains vertices a_1 and a_2 . Then there can be at most one vertex in B that is adjacent to both a_1 and a_2 in T , by the lack of 4-cycles. So $c(A') \geq n/2 - 1 \geq |A'|$. Finally, assume $A' = A$. Since every vertex in B has less than $|A|$ neighbors in T , the set A dominates all of B in the complement; that is, $c(A) \geq n/2$. It follows from Hall's Theorem that there exists a perfect matching P in $\bar{T}$ that joins A with B .

Assume second that in T there is a vertex v of degree at least $n/2$. Let w be a neighbor of v of largest degree, necessarily more than 1, since T is not a star. Construct set A to be $n/2 - 1$ neighbors of v excluding w , along with one neighbor of w (that is not v). It can readily be checked that this set A satisfies the conditions of Hall's Theorem: no vertex is adjacent to all of A in T , and no vertex of A is adjacent to all of $T - A$ in T , and the rest of the argument is the same as above. $\square$

The value in Theorem 11 is obviously best possible since there are trees with independence number half their order. Indeed, for the tree Q obtained from the star by subdividing exactly one edge, there is up to symmetry a unique perfect

matching P in the complement, and the independence number of $Q + P$ is half its order.

In contrast, the answer to Problem MM is $\lceil n/3 \rceil$. For any tree T and perfect matching P in $\overline{T}$, the graph $T + P$ is 3-colorable (since any subgraph has minimum degree at most 2). Thus $T + P$ is guaranteed to have an independent set of size at least $\lceil n/3 \rceil$. For the path and many other trees when n is a multiple of 3, one can readily choose a perfect matching such that the result has a spanning subgraph consisting of triangles, whence the bound is best possible.

6 Upper Domination and Upper Total Domination

Finally we consider the upper version of domination parameters (noting that independence number plays that role for independent domination). Recall that $\Gamma(G)$ denotes the maximum size of a minimal dominating set of G , and $\Gamma_t(G)$ denotes the maximum size of a minimal total dominating set of G . For (ordinary) upper domination, we need the following simple bound involving the matching number, which seems not to have been noticed before. (A somewhat related result is given in [14] where it is shown that every nonempty regular graph has upper domination number at most half its order.) Let $\alpha'(G)$ denote the matching number of G .

Theorem 12 *For any graph G of order n , $\Gamma(G) + \alpha'(G) \leq n$. In particular, if G has a perfect matching then $\Gamma(G) \leq n/2$.*

PROOF. Fix a maximum matching M of G , and let X be the set of vertices not covered by M . Consider any minimal dominating set S of G and the subgraph $G[S]$ induced by S . If vertex $v \in S$ is not isolated in $G[S]$, then it is guaranteed to have an external private neighbor (that is, a vertex v' whose only neighbor in S is v). Let v' be one such external private neighbor. If vertex $v \in S$ is isolated but covered by M , then let v' be its neighbor joined by the edge of M . Note that no v' is in S . Indeed, all of S and the v' are distinct. It follows that $|S| + (|S| - |X|) \leq n$. Since $|X| = n - 2\alpha'(G)$, rearranged this inequality gives the desired bound. $\square$

It thus follows:

Theorem 13 *If T is a tree of even order n that is not a star and P a perfect matching in $\overline{T}$, then*

$$\Gamma(T + P) \leq n/2,$$

and this is best possible.

Equality in Theorem 13 follows from the sharpness of the independence number results (and recalling that $\Gamma(G) \geq \alpha(G)$). On the other hand, the bound for upper total domination is much larger.

Theorem 14 *If T is a tree of even order n that is not a star and P a perfect matching in $\overline{T}$, then*

$$\Gamma_t(T + P) \leq n - 2,$$

and this is best possible.

PROOF. Note that the graph $T + P$ has minimum degree at least 2. It is known that there are graphs with minimum degree 2 that have upper total domination number one less than their order. However, we observe here that such graphs do not exist for even order.

Consider any graph G with minimum degree at least 2, and suppose S is a minimal total dominating set of G of size $n - 1$. Consider the subgraph $G[S]$ induced by S . Since $n - 1$ is odd, the graph $G[S]$ must contain a component C of odd order, necessarily at least 3. Let w be a non-cut-vertex of this component. Then we claim $S - \{w\}$ is a total dominating set of G . Both w and the vertices of S adjacent to w are still dominated by $S - \{w\}$, and the unique vertex outside S had at least two neighbors in S , and so is still dominated. This contradicts the minimality of S . $\square$

The simplest extremal graph for Theorem 14 is the tree Q formed from the star by subdividing one edge. As noted before, up to isomorphism there is a unique perfect matching P in the complement of Q . Further, the set X of neighbors of the center in Q is a minimal total dominating set of the resultant $Q + P$. This example also shows that the answer to Problem MM is $n - 2$ too.

7 Future Work

The most obvious open question is: what is the maximum value over all trees of the minimum domination number achieved by adding a perfect matching. In particular, does every tree have a matching where the resultant domination number is at most $4n/15$? It would also seem of interest to consider analogous questions for other domination parameters or other graph families.

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